1) a)

$$\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}=\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}$$

So $(2, 3, -1)$

1) b)

$$\begin{pmatrix}2 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & -3\end{pmatrix}\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}=\begin{pmatrix}4\\ 15\\ 3\end{pmatrix}$$

So $(4, 15, 3)$

1) c)

$$\begin{pmatrix}1 & 0 & 1\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}=\begin{pmatrix}1\\ 3\\ -1\end{pmatrix}$$

So $(1, 3, -1)$

1) d)

$$\begin{pmatrix}1 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 1\end{pmatrix}\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}=\begin{pmatrix}1\\ 3\\ 1\end{pmatrix}$$

So $(1, 3, 1)$

1) e)

$$\begin{pmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{pmatrix}\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}=\begin{pmatrix}4\\ 4\\ 4\end{pmatrix}$$

So $(4, 4, 4)$

1) f)

$$\begin{pmatrix}0 & 2 & 0\\ -3 & -5 & 4\\ 0 & 3 & 0\end{pmatrix}\begin{pmatrix}2\\ 3\\ -1\end{pmatrix}=\begin{pmatrix}6\\ -25\\ 9\end{pmatrix}$$

So $(6, -25, 9)$

2) a)

This is the identity matrix. All points get mapped to themselves.

2) b)

This is the zero matrix. All points get mapped to the origin, $(0, 0, 0)$

3)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a+4b+2c & a+b & a+3b+c \\ d+4e+2f & d+e & d+3e+f \\ g+4h+2i & g+h & g+3h+i \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}$$

$a + 4b + 2c = 11$ (1)

$a + b = 3$ (2)

$a + 3b + c = 8$ (3)

$\therefore a = 1, b = 2, c = 1$

$d + 4e + 2f = 6$ (4)

$d + e = 1$ (5)

$d + 3e + f = 4$ (6)

$\therefore d = 0\ , e = 1, f = 1$

$g + 4h + 2i = 8$ (7)

$g + h = 3$ (8)

$g + 3h + i = 6$ (9)

$\therefore g = 2\ , h = 1, i = 1$

So $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}$

Alternative Method (using inverse matrices):

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}^{-1}$$

$$\Rightarrow \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 11 & 3 & 8 \\ 6 & 1 & 4 \\ 8 & 3 & 6 \end{pmatrix}\begin{pmatrix} 1 & -1 & 2 \\ 2 & -1 & 1 \\ -2 & 2 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}$$

4)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a+b+c & 2a+b+2c & 3a+b+c \\ d+e+f & 2d+e+2f & 3d+e+f \\ g+h+i & 2g+h+2i & 3g+h+i \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}$$

$a + b + c = 2$ (1)

$2a + b + 2c = 1$ (2)

$3a + b + c = 0$ (3)

$\therefore a = -1, b = 3, c = 0$

$d + e + f = 3$ (4)

$2d + e + 2f = 2$ (5)

$3d + e + f = 0$ (6)

$\therefore d = -\frac{3}{2}, e = 4, f = \frac{1}{2}$

$g + h + i = 4$ (7)

$2g + h + 2i = 1$ (8)

$3g + h + i = 1$ (9)

$\therefore g = -\frac{3}{2}, h = 7, i = -\frac{3}{2}$

So $\begin{pmatrix} -1 & 3 & 0 \\ -\frac{3}{2} & 4 & \frac{1}{2} \\ -\frac{3}{2} & 7 & -\frac{3}{2} \end{pmatrix}$

Alternative Method (using inverse matrices):

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}^{-1}$$

$$\Rightarrow \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 4 & 1 & 1 \end{pmatrix}\begin{pmatrix} -\frac{1}{2} & 2 & -\frac{1}{2} \\ 0 & -1 & 1 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & 3 & 0 \\ -\frac{3}{2} & 4 & \frac{1}{2} \\ -\frac{3}{2} & 7 & -\frac{3}{2} \end{pmatrix}$$

5)

$$\begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix}\begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 5a+2b-c & 5d+2e-f & 5g+2h-i \\ -a-c & -d-f & -g-i \\ a+2b+5c & d+2e+5f & g+2h+5i \end{pmatrix} = \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$5a + 2b - c = 14$ (1)

$-a - c = 0$ (2)

$a + 2b + 5c = -16$ (3)

$\therefore a = 3, b = -2, c = -3$

$5d + 2e - f = -7$ (4)

$-d - f = -1$ (5)

$d + 2e + 5f = 19$ (6)

$\therefore d = -2, e = 3, f = 3$

$5g + 2h - i = -21$ (7)

$-g - i = 1$ (8)

$g + 2h + 5i = 3$ (9)

$\therefore g = -3, h = -2, i = 2$

So $\begin{pmatrix} 3 & -2 & -3 \\ -2 & 3 & -2 \\ -3 & 3 & 2 \end{pmatrix}$

Alternative Method (using inverse matrices):

$$\begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix}\begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix}^{-1}\begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} = \begin{pmatrix} \frac{1}{10} & -\frac{3}{5} & -\frac{1}{10} \\ \frac{1}{5} & \frac{13}{10} & \frac{3}{10} \\ -\frac{1}{10} & -\frac{2}{5} & \frac{1}{10} \end{pmatrix}\begin{pmatrix} 14 & -7 & -21 \\ 0 & -1 & 1 \\ -16 & 19 & 3 \end{pmatrix} = \begin{pmatrix} 3 & -2 & -3 \\ -2 & 3 & -2 \\ -3 & 3 & 2 \end{pmatrix}$$

6)

$$\begin{pmatrix}2 & 1 & 2\\ 2 & 2 & 2\\ 1 & 2 & 1\end{pmatrix}\begin{pmatrix}a & d & g\\ b & e & h\\ c & f & i\end{pmatrix}=\begin{pmatrix}-6 & -3 & 6\\ -8 & -2 & 6\\ -6 & 0 & 3\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}2a+b+2c & 2d+e+2f & 2g+h+2i\\ 2a+2b+2c & 2d+2e+2f & 2g+2h+2i\\ a+2b+c & d+2e+f & g+2h+i\end{pmatrix}=\begin{pmatrix}-6 & -3 & 6\\ -8 & -2 & 6\\ -6 & 0 & 3\end{pmatrix}$$

$2a+b+2c=-6$ (1)

$2a+2b+2c=-8$ (2)

$a+2b+c=-6$ (3)

$\therefore a+c=-2,\ b=-2$

$2d+e+2f=-3$ (4)

$2d+2e+2f=-2$ (5)

$d+2e+f=0$ (6)

$\therefore d+f=-2,\ e=1$

$2g+h+2i=6$ (7)

$2g+2h+2i=6$ (8)

$g+2h+i=3$ (9)

$\therefore g+i=3,\ h=0$

So $b=-2,\ e=1,\ h=0,\ c=-2-a,\ f=-2-d,\ i=3-g$

$$\therefore\begin{pmatrix}a & d & g\\ -2 & 1 & 0\\ -2-a & -2-d & 3-g\end{pmatrix}$$

7) a)

Stretch, parallel to the x-axis, scale factor 5

7) b)

Stretch, parallel to the y-axis, scale factor 4

7) c)

Stretch, parallel to the z-axis, scale factor 6

7) d)

Stretch, parallel to the x-axis, scale factor 7, and a Stretch, parallel to the z-axis, scale factor 2

7) e)

Enlargement, centred at the origin, scale factor 3

Alternative Method:

Stretch, parallel to the x-axis, scale factor 3, and a Stretch, parallel to the y-axis, scale factor 3, and a Stretch, parallel to the z-axis, scale factor 3

8)

The matrix represents a Stretch, parallel to the x-axis, scale factor 2, and a Stretch, parallel to the y-axis, scale factor 3. So the image will be $2 \times 3 \times$ the volume of the original shape. So the volume of the shape's image is $2 \times 3 \times 4 = 24$ units^3

9)

The matrix represents an Enlargement, centred at the origin, scale factor 5. So the image will be $5^3 \times$ the volume of the original shape. So the volume of the shape's image is $5^3 \times 4 = 500$ units^3

10)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 10 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 20 & 0 & 0 \\ 0 & 0 & 30 \\ 0 & 0 & 0 \end{pmatrix}$$

The coordinates of the vertices of triangle T' are $(20, 0, 0)$, $(0, 0, 0)$ and $(0, 30, 0)$

The area of the triangle is $= \frac{1}{2} \times 20 \times 30 = 300$ units^2

11)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 8 & 0 & 0 \\ 0 & 0 & -4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 24 & 0 & 0 \\ 0 & 0 & -8 \end{pmatrix}$$

The coordinates of the vertices of triangle T' are $(0, 24, 0)$, $(0, 0, 0)$ and $(0, 0, -8)$

The area of the triangle is $= \frac{1}{2} \times 24 \times 8 = 96$ units^2

12) a)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

12) b)

$$\begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

12) c)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

12) d)

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

12) e)

$$\begin{pmatrix} \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{8} \end{pmatrix}$$

13)

Area of K is $\frac{7\times4}{2} = 14$ units^2

$$\begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix}\begin{pmatrix} 0 & -2 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ -4 & 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2k & 0 & 2k \\ 0 & 0 & 0 & 0 \\ -4k & 0 & 3k & 0 \end{pmatrix}$$

Area of $K' = \frac{7a\times4a}{2} = 14a^2$

$14a^2 = 3 \times 14 \Rightarrow a^2 = 3 \Rightarrow a = \sqrt{3}$

So the vertices of K' have coordinates $(0, 0, -4\sqrt{3})$, $(-2\sqrt{3}, 0, 0)$, $(0, 0, 3\sqrt{3})$ and $(2\sqrt{3}, 0, 0)$

14) a)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

14) b)

$$\begin{pmatrix}4 & 0 & 0\\ 0 & 8 & 0\\ 0 & 0 & 9\end{pmatrix}\begin{pmatrix}3m+n\\ n-m\\ m+2n\end{pmatrix}=\begin{pmatrix}16m-n\\ -2n\\ 21m+3n\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}12m+4n\\ 8n-8m\\ 9m+18n\end{pmatrix}=\begin{pmatrix}16m-n\\ -2n\\ 21m+3n\end{pmatrix}$$

$12m+4n=16m-n \Rightarrow 4m-5n=0$ (1)

$8n-8m=-2n \Rightarrow 8m-10n=0 \Rightarrow 4m-5n=0$ (1)

$9m+18n=21m+3n \Rightarrow 12m-15n=0 \Rightarrow 4m-5n=0$ (1)

$m+n=9$ (2)

$\therefore m=5, n=4$

15) a)

Reflection in the plane $y=0$

15) b)

Reflection in the plane $x=0$

15) c)

Reflection in the plane $z=0$

16)

The matrix represents a Reflection in the plane $z=0$. So the image will have the same volume as the original shape. So the volume of the shape's image is 4 units^3

17)

$$\begin{pmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}2 & 1 & 3\\ 1 & 4 & 1\\ 1 & 1 & 4\end{pmatrix}=\begin{pmatrix}2 & 1 & 3\\ -1 & -4 & -1\\ 1 & 1 & 4\end{pmatrix}$$

So the vertices of T' have coordinates $(2,-1,1)$, $(1,-4,1)$ and $(3,-1,4)$

18)

$$\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}3 & 3 & 0 & -2\\ 0 & 0 & 3 & 5\\ 0 & 5 & 0 & 5\end{pmatrix}=\begin{pmatrix}-3 & -3 & 0 & 2\\ 0 & 0 & 3 & 5\\ 0 & 5 & 0 & 5\end{pmatrix}$$

So the vertices of X' have coordinates $(-3,0,0)$, $(-3,0,5)$, $(0,3,0)$ and $(2,5,5)$

19) a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

19) b)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

19) c)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

19) d)

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

19) e)

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

19) f)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

20)

$\mathbf{A}$ represents a reflection so $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$\therefore 1 - 2k^2(2k + 3) = -1 \Rightarrow 2 - 4k^3 - 6k^2 = 0 \Rightarrow 4k^3 + 6k^2 - 2 = 0$

$\Rightarrow 2k^3 + 3k^2 - 1 = 0 \Rightarrow (2k - 1)(k + 1)^2 = 0 \Rightarrow k = \frac{1}{2}$ or $k = -1$

21) a)

A rotation by 90°, anticlockwise about the x-axis

21) b)

A rotation by 270°, anticlockwise about the y-axis

21) c)
A rotation by $180°$ about the x-axis

21) d)
A rotation by $180°$ about the z-axis

21) e)
A rotation by $90°$, anticlockwise about the y-axis

21) f)
A rotation by $270°$, anticlockwise about the x-axis

21) g)
A rotation by $270°$, anticlockwise about the z-axis

21) h)
A rotation by $180°$ about the y-axis

21) i)
A rotation by $90°$, anticlockwise about the z-axis

22)
The matrix represents a Rotation by $90°$, anticlockwise about the y-axis. So the image will have the same volume as the original shape. So the volume of the shape's image is 4 units^3

23)
$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 7 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \\ 1 \end{pmatrix}$$
So the coordinates of B are $(2, 7, 1)$

24)
$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \\ 5 \end{pmatrix}$$
So the coordinates of A are $(-3, 4, 5)$

25)

$$\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & -1\end{pmatrix}\begin{pmatrix}-1 & 2 & 4\\ 1 & 5 & -3\\ 3 & 1 & -2\end{pmatrix}=\begin{pmatrix}1 & -2 & -4\\ 1 & 5 & -3\\ -3 & -1 & 2\end{pmatrix}$$

So the coordinates of the vertices of T' are $(1,1,-3)$, $(-2,5,-1)$ and $(-4,-3,2)$

26)

$$\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & -1 & 0\end{pmatrix}\begin{pmatrix}p\\ q\\ r\end{pmatrix}=\begin{pmatrix}a\\ b\\ c\end{pmatrix}\Rightarrow\begin{pmatrix}p\\ r\\ -q\end{pmatrix}=\begin{pmatrix}a\\ b\\ c\end{pmatrix}$$

So the coordinates of P are $(a,-c,b)$

27) a)

$$\begin{pmatrix}1 & 0 & 0\\ 0 & \cos 45° & -\sin 45°\\ 0 & \sin 45° & \cos 45°\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{pmatrix}$$

27) b)

$$\begin{pmatrix}\cos 135° & 0 & \sin 135°\\ 0 & 1 & 0\\ -\sin 135° & 0 & \cos 135°\end{pmatrix}=\begin{pmatrix}-\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2}\\ 0 & 1 & 0\\ -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2}\end{pmatrix}$$

27) c)

$$\begin{pmatrix}\cos 315° & -\sin 315° & 0\\ \sin 315° & \cos 315° & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0\\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0\\ 0 & 0 & 1\end{pmatrix}$$

28) a)

$$\begin{pmatrix}1 & 0 & 0\\ 0 & \cos 30° & -\sin 30°\\ 0 & \sin 30° & \cos 30°\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2}\\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}$$

28) b)

$$\begin{pmatrix} \cos 120° & 0 & \sin 120° \\ 0 & 1 & 0 \\ -\sin 120° & 0 & \cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{pmatrix}$$

28) c)

$$\begin{pmatrix} \cos 150° & -\sin 150° & 0 \\ \sin 150° & \cos 150° & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

28) d)

A rotation by $60°$, clockwise about the x-axis, is the same as a rotation by $300°$, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 300° & -\sin 300° \\ 0 & \sin 300° & \cos 300° \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

28) e)

A rotation by $225°$, clockwise about the y-axis, is the same as a rotation by $135°$, anticlockwise about the y-axis.

$$\begin{pmatrix} \cos 135° & 0 & \sin 135° \\ 0 & 1 & 0 \\ -\sin 135° & 0 & \cos 135° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

28) f)

A rotation by $330°$, clockwise about the z-axis, is the same as a rotation by $30°$, anticlockwise about the z-axis.

$$\begin{pmatrix} \cos 30° & -\sin 30° & 0 \\ \sin 30° & \cos 30° & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

29)

$$\begin{pmatrix} \cos 35° & 0 & \sin 35° \\ 0 & 1 & 0 \\ -\sin 35° & 0 & \cos 35° \end{pmatrix} = \begin{pmatrix} 0.819 & 0 & 0.574 \\ 0 & 1 & 0 \\ -0.574 & 0 & 0.819 \end{pmatrix} \text{ (3 SF)}$$

30)

A rotation by 102°, clockwise about the x-axis, is the same as a rotation by 258°, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 258° & -\sin 258° \\ 0 & \sin 258° & \cos 258° \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.208 & 0.978 \\ 0 & -0.978 & -0.208 \end{pmatrix} \text{ (3 SF)}$$

31)

$$\begin{pmatrix} \cos 45° & -\sin 45° & 0 \\ \sin 45° & \cos 45° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & 5 \\ 9 & 0 & -4 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & 5 \\ 9 & 0 & -4 \end{pmatrix} = \begin{pmatrix} \frac{3\sqrt{2}}{2} & -\frac{3\sqrt{2}}{2} & -\frac{5\sqrt{2}}{2} \\ \frac{3\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \frac{5\sqrt{2}}{2} \\ 9 & 0 & -4 \end{pmatrix}$$

So the coordinates of the vertices of T' are $\left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}, 9\right)$, $\left(-\frac{3\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0\right)$ and $\left(-\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2}, -4\right)$

32)

A rotation by 135°, clockwise about the x-axis, is the same as a rotation by 225°, anticlockwise about the x-axis.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 225° & -\sin 225° \\ 0 & \sin 225° & \cos 225° \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 3 & 0 & 0 & 3 \\ -1 & -1 & 2 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 3 & 0 & 0 & 3 \\ -1 & -1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 0 \\ -2\sqrt{2} & -\frac{\sqrt{2}}{2} & \sqrt{2} & -\frac{\sqrt{2}}{2} \\ -\sqrt{2} & \frac{\sqrt{2}}{2} & -\sqrt{2} & -\frac{5\sqrt{2}}{2} \end{pmatrix}$$

So the coordinates of the vertices of R' are $(0, -2\sqrt{2}, -\sqrt{2})$, $\left(2, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $(2, \sqrt{2}, -\sqrt{2})$ and $\left(0, -\frac{\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2}\right)$

33)

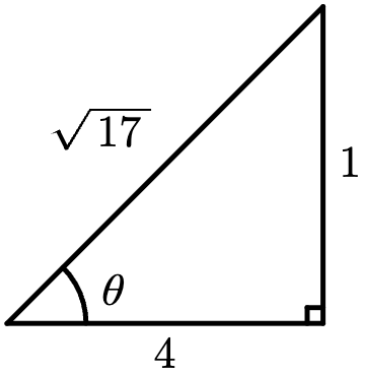
As the rotation is about the z-axis, the z-coordinate of A will be the same as the z-coordinate of A'

$$\begin{pmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}16\\ -64\\ 4\end{pmatrix}=\begin{pmatrix}0\\ k\\ 4\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}16\cos\theta+64\sin\theta\\ 16\sin\theta-64\cos\theta\\ 4\end{pmatrix}=\begin{pmatrix}0\\ k\\ 4\end{pmatrix}$$

$16\cos\theta+64\sin\theta=0\Rightarrow 16+\frac{64\sin\theta}{\cos\theta}=0\Rightarrow 16+64\tan\theta=0$

$\Rightarrow\tan\theta=-\frac{1}{4}$



As θ is obtuse, we must have $\sin\theta=\frac{1}{\sqrt{17}}$ and $\cos\theta=-\frac{4}{\sqrt{17}}$

So $k=16\left(\frac{1}{\sqrt{17}}\right)-64\left(-\frac{4}{\sqrt{17}}\right)=16\sqrt{17}$

$\therefore A'\left(0,16\sqrt{17},4\right)$

34)

A rotation by $45°$, clockwise about the x-axis, is the same as a rotation by $315°$, anticlockwise about the x-axis.

$$\begin{pmatrix}1 & 0 & 0\\ 0 & \cos 315° & -\sin 315°\\ 0 & \sin 315° & \cos 315°\end{pmatrix}\begin{pmatrix}1 & 0 & 2\\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\\ k\sqrt{2} & -\frac{3\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\end{pmatrix}=\begin{pmatrix}1 & 0 & 2\\ 2 & -1 & 0\\ 2 & -2 & -1\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 0 & 0\\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{pmatrix}\begin{pmatrix}1 & 0 & 2\\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\\ k\sqrt{2} & -\frac{3\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\end{pmatrix}=\begin{pmatrix}1 & 0 & 2\\ 2 & -1 & 0\\ 2 & -2 & -1\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 0 & 2\\ k & -1 & 0\\ k & -2 & -1\end{pmatrix}=\begin{pmatrix}1 & 0 & 2\\ 2 & -1 & 0\\ 2 & -2 & -1\end{pmatrix}$$

$\therefore k=2$

35) a)

$\cos\theta = -\frac{1}{2} \Rightarrow \theta = \{120°, 240°\}$

$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{60°, 120°\}$

So it is a rotation by $120°$, anticlockwise about the x-axis.

35) b)

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30°, 330°\}$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = \{30°, 150°\}$

So it is a rotation by $30°$, anticlockwise about the z-axis.

35) c)

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = \{225°, 315°\}$

So it is a rotation by $315°$, anticlockwise about the y-axis.

35) d)

$\cos\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{15°, 345°\}$

$\sin\theta = \frac{\sqrt{2}-\sqrt{6}}{4} \Rightarrow \theta = \{195°, 345°\}$

So it is a rotation by $345°$, anticlockwise about the y-axis.

35) e)

$\cos\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{15°, 345°\}$

$\sin\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{15°, 165°\}$

So it is a rotation by $15°$, anticlockwise about the z-axis.

36)

$\cos\theta = \frac{1}{3} \Rightarrow \theta = \{70.5°, 289°\}$ (3 SF)

$\sin\theta = \frac{2\sqrt{2}}{3} \Rightarrow \theta = \{70.5°, 109°\}$ (3 SF)

So $\theta = 70.5°$ (3 SF)

37)

$\cos\theta = \frac{\sqrt{7}}{4} \Rightarrow \theta = \{48.6°, 311°\}$ (3 SF)

$\sin\theta = \frac{3}{4} \Rightarrow \theta = \{48.6°, 131°\}$ (3 SF)

So $\theta = 48.6°$ (3 SF)

38)

$\cos\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{75°, 285°\}$

$\sin\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{75°, 105°\}$

So $\theta = 75°$

$75n = 360k$

$3 \times 5^2 \times n = 2^3 \times 3^2 \times 5 \times k \Rightarrow n = 2^3 \times 3 \times 5^{-1} \times k$

Smallest integer value for n when $k = 5$, so $n = 24$

39)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} -3 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ -7 \\ -4 \end{pmatrix} \Rightarrow \begin{pmatrix} -3 \\ 4\cos\theta - 7\sin\theta \\ 4\sin\theta + 7\cos\theta \end{pmatrix} = \begin{pmatrix} -3 \\ -7 \\ -4 \end{pmatrix}$$

$4\cos\theta - 7\sin\theta = -7$ (1)

$4\sin\theta + 7\cos\theta = -4$ (2)

$\therefore \cos\theta = -\frac{56}{65}$ and $\sin\theta = \frac{33}{65}$

$\cos\theta = -\frac{56}{65} \Rightarrow \theta = \{149°, 211°\}$ (3 SF)

$\sin\theta = \frac{33}{65} \Rightarrow \theta = \{30.5°, 149°\}$ (3 SF)

So $\theta = 149°$ (3 SF)

40)

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}\begin{pmatrix} 1 \\ 8 \\ -6 \end{pmatrix} = \begin{pmatrix} -6 \\ 8 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} \cos\theta - 6\sin\theta \\ 8 \\ -\sin\theta - 6\cos\theta \end{pmatrix} = \begin{pmatrix} -6 \\ 8 \\ 1 \end{pmatrix}$$

$\cos\theta - 6\sin\theta = -6$ (1)

$-\sin\theta - 6\cos\theta = 1$ (2)

$\therefore \cos\theta = -\frac{12}{37}$ and $\sin\theta = \frac{35}{37}$

$\cos\theta = -\frac{12}{37} \Rightarrow \theta = \{109°, 251°\}$ (3 SF)

$\sin\theta = \frac{35}{37} \Rightarrow \theta = \{71.1°, 109°\}$ (3 SF)

So $\theta = 109°$ (3 SF)

41)

$$\begin{pmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}4\\8\\2\end{pmatrix}=\begin{pmatrix}-8\\-4\\2\end{pmatrix}\Rightarrow\begin{pmatrix}4\cos\theta-8\sin\theta\\4\sin\theta+8\cos\theta\\2\end{pmatrix}=\begin{pmatrix}-8\\-4\\2\end{pmatrix}$$

$4\cos\theta-8\sin\theta=-8$ (1)

$4\sin\theta+8\cos\theta=-4$ (2)

$\therefore \cos\theta=-\frac{4}{5}$ and $\sin\theta=\frac{3}{5}$

$\cos\theta=-\frac{4}{5}\Rightarrow\theta=\{143°,217°\}$ (3 SF)

$\sin\theta=\frac{3}{5}\Rightarrow\theta=\{36.9°,143°\}$ (3 SF)

So $\theta=143°$ (3 SF)

42)

$$\begin{pmatrix}\cos\theta & 0 & \sin\theta\\ 0 & 1 & 0\\ -\sin\theta & 0 & \cos\theta\end{pmatrix}\begin{pmatrix}5\\-5\\-6\end{pmatrix}=\begin{pmatrix}6\\-5\\-5\end{pmatrix}\Rightarrow\begin{pmatrix}5\cos\theta-6\sin\theta\\-5\\-5\sin\theta-6\cos\theta\end{pmatrix}=\begin{pmatrix}6\\-5\\-5\end{pmatrix}$$

$5\cos\theta-6\sin\theta=6$ (1)

$-5\sin\theta-6\cos\theta=-5$ (2)

$\therefore \cos\theta=\frac{60}{61}$ and $\sin\theta=-\frac{11}{61}$

$\cos\theta=\frac{60}{61}\Rightarrow\theta=\{10.4°,350°\}$ (3 SF)

$\sin\theta=-\frac{11}{61}\Rightarrow\theta=\{190°,350°\}$ (3 SF)

So $\theta=350°$ (3 SF) about the y-axis

43) a)

$$\mathbf{BA}=\begin{pmatrix}2&0&0\\0&1&-1\\0&1&-1\end{pmatrix}\begin{pmatrix}-1&0&1\\0&2&0\\1&0&-1\end{pmatrix}=\begin{pmatrix}-2&0&2\\-1&2&1\\-1&2&1\end{pmatrix}$$

43) b)

$$\mathbf{AB}=\begin{pmatrix}-1&0&1\\0&2&0\\1&0&-1\end{pmatrix}\begin{pmatrix}2&0&0\\0&1&-1\\0&1&-1\end{pmatrix}=\begin{pmatrix}-2&1&-1\\0&2&-2\\2&-1&1\end{pmatrix}$$

44)

$$\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}\begin{pmatrix}-1&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}-1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$$

So this is a Rotation by 180° about the y-axis.

45) a)
Rotation by 90°, anticlockwise about the x-axis

45) b)
$\mathbf{A}^2$ is a rotation by 180°, anticlockwise about the x-axis
$\mathbf{A}^3$ is a rotation by 270°, anticlockwise about the x-axis
$\mathbf{A}^4$ is a rotation by 360°, anticlockwise about the x-axis
$\mathbf{A}^5$ is a rotation by 90°, anticlockwise about the x-axis
$\mathbf{A}^6$ is a rotation by 180°, anticlockwise about the x-axis
$\mathbf{A}^7$ is a rotation by 270°, anticlockwise about the x-axis
$\mathbf{A}^8$ is a rotation by 360°, anticlockwise about the x-axis
…
$\frac{2025}{4} = 506$ r 1
So $\mathbf{A}^{2025}$ is a rotation by 90°, anticlockwise about the x-axis

46) a)
Rotation by 270°, anticlockwise about the z-axis

46) b)
$\mathbf{A}^2$ is a rotation by 180°, anticlockwise about the z-axis
$\mathbf{A}^3$ is a rotation by 90°, anticlockwise about the z-axis
$\mathbf{A}^4$ is a rotation by 360°, anticlockwise about the z-axis
$\mathbf{A}^5$ is a rotation by 270°, anticlockwise about the z-axis
$\mathbf{A}^6$ is a rotation by 180°, anticlockwise about the z-axis
$\mathbf{A}^7$ is a rotation by 90°, anticlockwise about the z-axis
$\mathbf{A}^8$ is a rotation by 360°, anticlockwise about the z-axis
…
$\frac{443}{4} = 110$ r 3
So $\mathbf{A}^{443}$ is a rotation by 90°, anticlockwise about the z-axis

47) a)
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{pmatrix}$$

47) b)

$$\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}-1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

47) c)

$$\begin{pmatrix}1 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & -1\\ 0 & 1 & 0\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & -5\\ 0 & 1 & 0\end{pmatrix}$$

47) d)

$$\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & -1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & -1\end{pmatrix}=\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

47) e)

$$\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}0 & 1 & 0\\ -1 & 0 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}0 & 1 & 0\\ -1 & 0 & 0\\ 0 & 0 & 1\end{pmatrix}$$

48)

$$\begin{pmatrix}-k & m & 0\\ \frac{4k}{m} & k & 0\\ 0 & 0 & -k\sqrt{3}\end{pmatrix}^2=\begin{pmatrix}12 & 0 & 0\\ 0 & 12 & 0\\ 0 & 0 & 12\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}k^2+4k & 0 & 0\\ 0 & 4k+k^2 & 0\\ 0 & 0 & 3k^2\end{pmatrix}=\begin{pmatrix}12 & 0 & 0\\ 0 & 12 & 0\\ 0 & 0 & 12\end{pmatrix}$$

$k^2+4k=12\Rightarrow k=-6$ or $k=2$

$3k^2=12\Rightarrow k=\pm 2$

$\therefore k=2$

49)

$$\begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix}\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 6a+3d+4g & 6b+3e+4h & 6c+3f+4i \\ -2a+2d+4g & -2b+2e+4h & -2c+2f+4i \\ 5a+4d+g & 5b+4e+h & 5c+4f+i \end{pmatrix} = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$6a+3d+4g=-5$ (1)

$-2a+2d+4g=0$ (2)

$5a+4d+g=0$ (3)

$\therefore a=-\frac{7}{9}, d=\frac{11}{9}, g=-1$

$6b+3e+4h=0$ (4)

$-2b+2e+4h=-5$ (5)

$5b+4e+h=0$ (6)

$\therefore b=\frac{13}{18}, e=-\frac{7}{9}, h=-\frac{1}{2}$

$6c+3f+4i=0$ (7)

$-2c+2f+4i=0$ (8)

$5c+4f+i=5$ (9)

$\therefore c=-\frac{2}{9}, f=\frac{16}{9}, i=-1$

$a=-\frac{7}{9}, b=\frac{13}{18}, c=-\frac{2}{9}, d=\frac{11}{9}, e=-\frac{7}{9}, f=\frac{16}{9}, g=-1, h=-\frac{1}{2}, i=-1$

So $\begin{pmatrix} -\frac{7}{9} & \frac{13}{18} & -\frac{2}{9} \\ \frac{11}{9} & -\frac{7}{9} & \frac{16}{9} \\ -1 & -\frac{1}{2} & -1 \end{pmatrix}$

Alternative Method (using inverse matrices):

$$\begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix}\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = \begin{pmatrix} \frac{7}{45} & -\frac{13}{90} & -\frac{2}{45} \\ -\frac{11}{45} & \frac{7}{45} & \frac{16}{45} \\ \frac{1}{5} & \frac{1}{10} & -\frac{1}{5} \end{pmatrix}\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{7}{9} & \frac{13}{18} & -\frac{2}{9} \\ \frac{11}{9} & -\frac{7}{9} & \frac{16}{9} \\ -1 & -\frac{1}{2} & -1 \end{pmatrix}$$

50) a)

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Stretch, parallel to the y-axis, scale factor 3, followed by an Enlargement, centred at the origin, scale factor 4

50) b)

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Reflection in the plane $y = 0$, followed by an Enlargement, centred at the origin, scale factor 6

50) c)

$$\begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A rotation by $180°$ about the z-axis, followed by an Enlargement, centred at the origin, scale factor 2

50) d)

$$\begin{pmatrix} 0 & 8 & 0 \\ -8 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A rotation by $270°$ anticlockwise about the z-axis, followed by an Enlargement, centred at the origin, scale factor 8